AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 24 · WORKBOOK

Prototype a Grounded Assistant

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Prototype a Grounded Assistant before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Block to Python Bridge

Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Tools and APIs

Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Retrieval Augmented Generation

Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Prototype a Grounded Assistant trace and record the first step you can explain without help.

Use W2: Trace the worked case for Capstone grounded assistant. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Prototype a Grounded Assistant explanation against the supplied facts before discussing it.

Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Prototype a Grounded Assistant target to a trace and a justified conclusion.

Build a paper assistant from notes A='Roots take up water' and B='Leaves use light'. Answer 'Which part takes up water?' and 'How tall is this plant?' with references or **fallback**.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Prototype a Grounded Assistant case and show the deciding evidence.

Approved notes say plants need light and roots take up water. Ask how roots help.

Specify the minimum grounded path.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A study assistant has a supplied passage about roots. A user asks a question about planets. What should the **prototype** do within a passage-only boundary?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 24 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Prototype a Grounded Assistant adds to Verified Study Buddy and keep the evidence.

Connect user interface, source store, retrieval, prompt, schema validation, and **evidence** display.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Build a paper assistant from notes A='Roots take up water' and B='Leaves use light'. Answer 'Which part takes up water?' and 'How tall is this plant?' with references or **fallback**.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Prototype a Grounded Assistant facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Prototype a Grounded Assistant facts and keep your reasoning.

Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Prototype a Grounded Assistant facts and keep your reasoning.

A report omits this qualification: A correct happy path does not establish **fallback**, access control or side-effect safety. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Prototype a Grounded Assistant facts and keep your reasoning.

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Prototype a Grounded Assistant facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Prototype a Grounded Assistant

> Build a minimal grounded study answer with checks between retrieval and display. Explain the labelled step before checking the changed condition.

Build a minimal grounded study answer with checks between retrieval and display.

> Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/24

Prototype a Grounded Assistant

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Retrieve evidence

PL3 directly supports the daily soil-check claim; PL2 supplies a different fact.

↓

2. Match question

Choose the activity's soil check, not a general watering recommendation.

↓

3. Validate draft

Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

CHANGE ONE THING

PL3 is absent while PL2 remains.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Build a minimal grounded study answer with checks between retrieval and display.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule.

1. Retrieve **evidence**

2. Match question

3. Validate draft

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Validate draft Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. Read the picture in order.

Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: PL3 is absent while PL2 remains. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Explain that the daily check is not specified in the available source.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Explain that the daily check is not specified in the available source. The remaining general water fact does not establish this activity’s schedule.

Repair a missing building block

Review Block to Python Bridge (ai-3/23). Goal: Translate a short paper calculation into a Python script. Short practice: Trace the changed list and identify where the result differs.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Tools and APIs (ai-3/21). Goal: Explain the **mechanism** and terminology of tools and api contracts. Short practice: Does a valid C7 ID permit the extra lending action?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

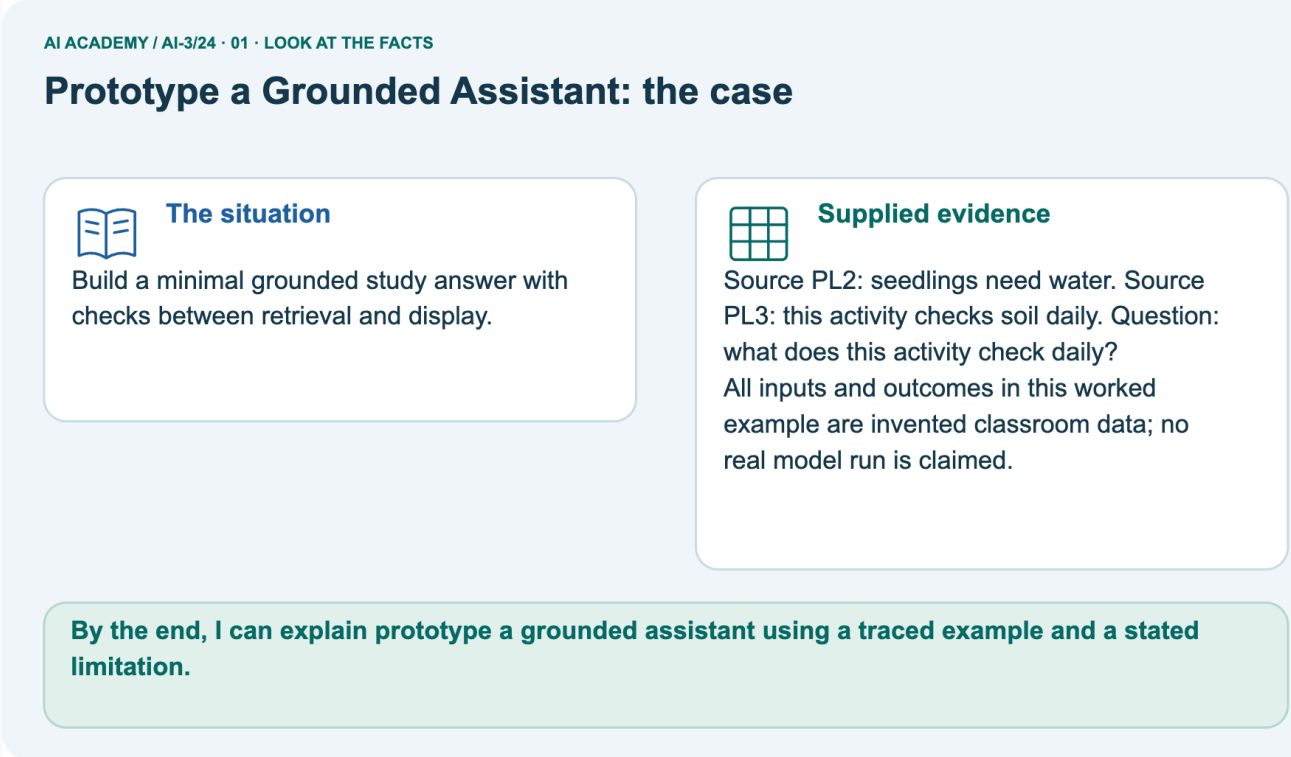
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Prototype a Grounded Assistant picture from its labels, then record a relevant limit.

01. Prototype a Grounded Assistant: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

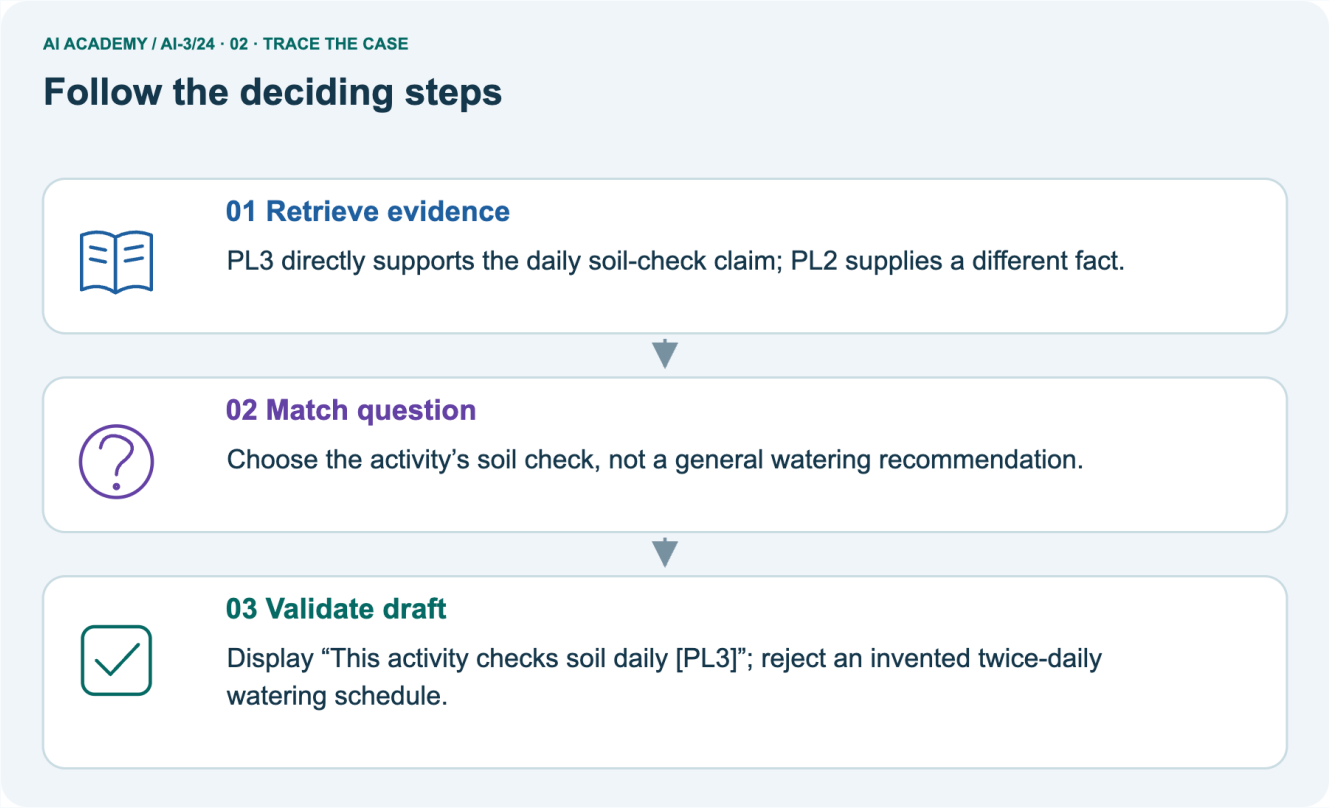
Build a minimal grounded study answer with checks between retrieval and display. Supplied facts: Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



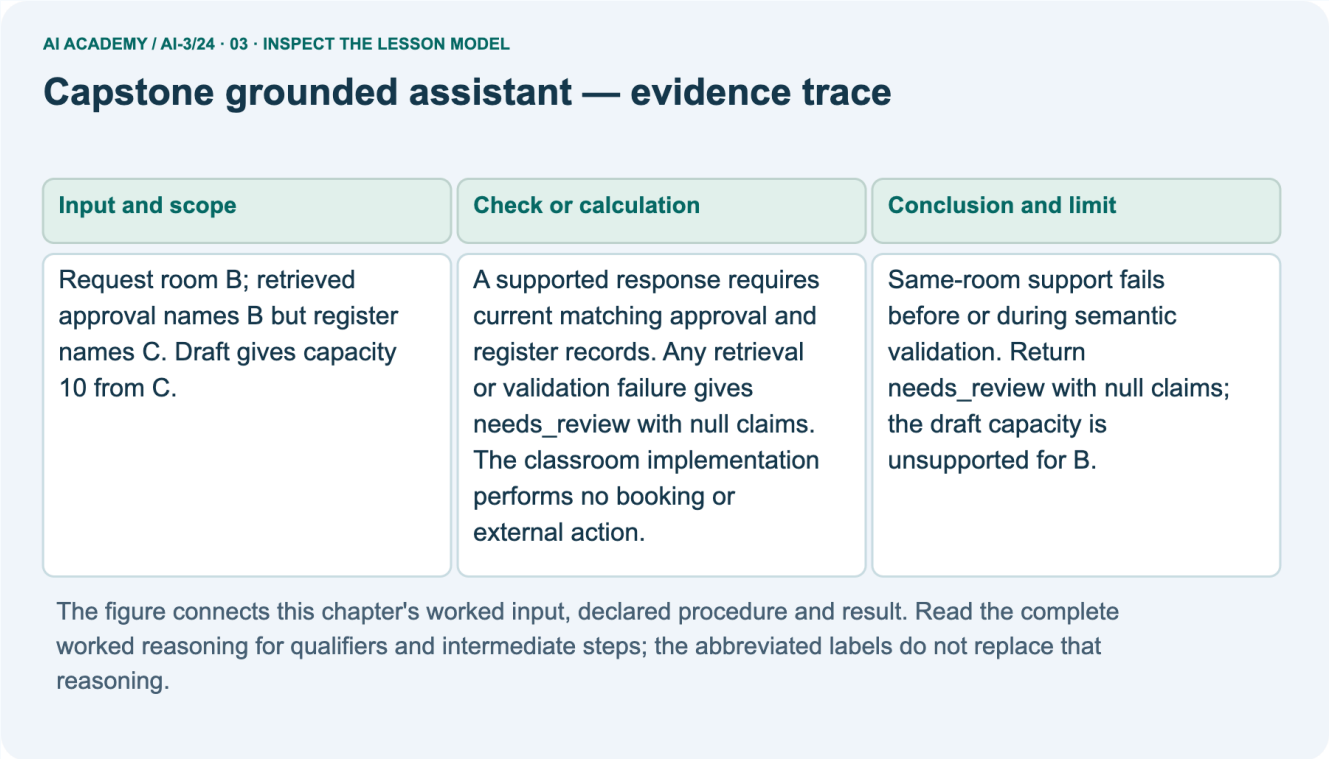
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve **evidence**: PL3 directly supports the daily soil-check claim; PL2 supplies a different fact. → Match question: Choose the activity's soil check, not a general watering recommendation. → Validate draft: Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone grounded assistant — evidence trace



AI ACADEMY / AI-3/24 · 03 · INSPECT THE LESSON MODEL

Capstone grounded assistant — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C.	A supported response requires current matching approval and register records. Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action.	Same-room support fails before or during semantic validation. Return needs_review with null claims; the draft capacity is unsupported for B.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone grounded assistant — **evidence** trace. Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. | A supported response requires current matching approval and register records.

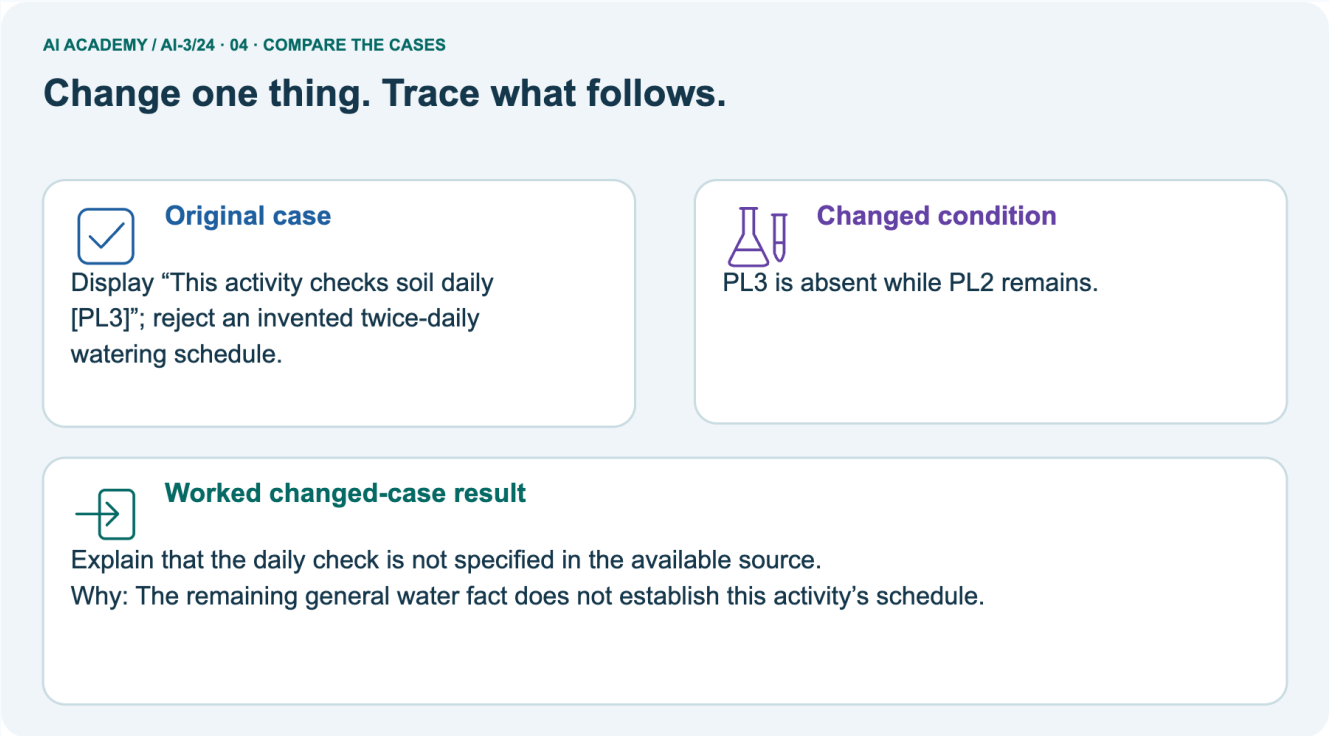
Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action. | Same-room support fails before or during semantic validation.

Return needs_review with null claims; the draft capacity is unsupported for B.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

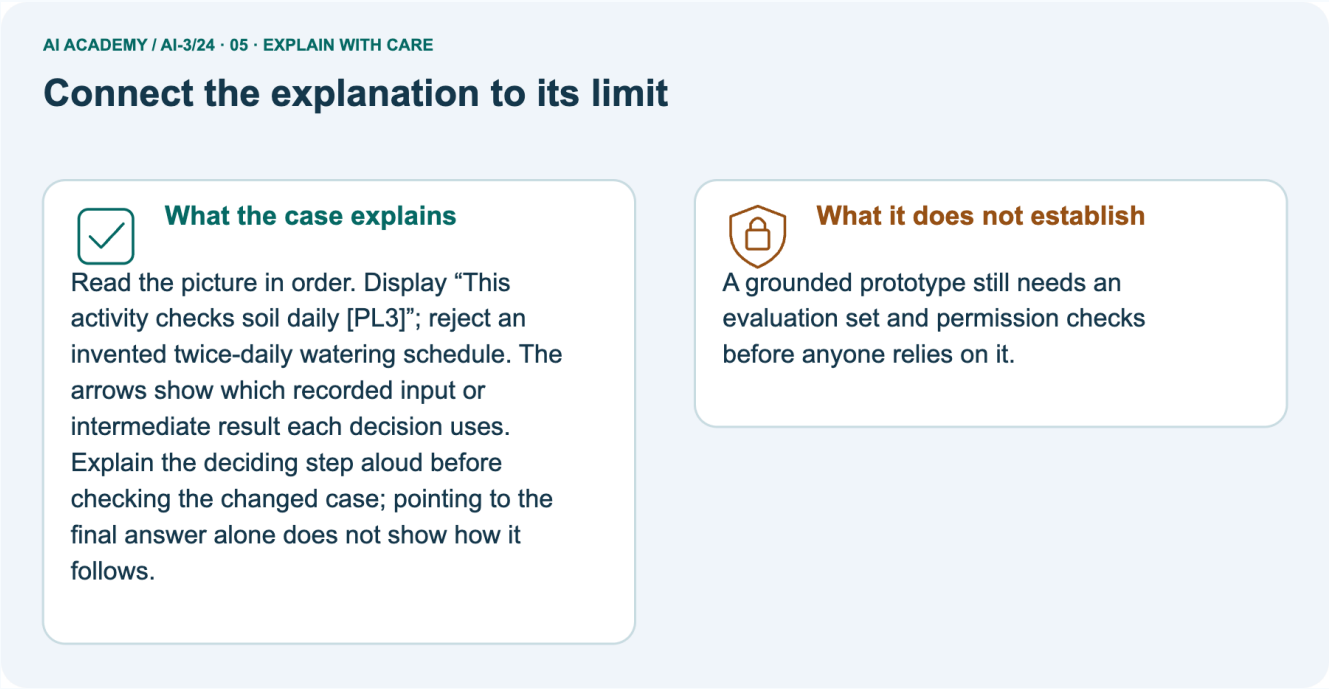
Original case: Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. Changed condition: PL3 is absent while PL2 remains.

Expected result: Explain that the daily check is not specified in the available source. Reason: The remaining general water fact does not establish this activity’s schedule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

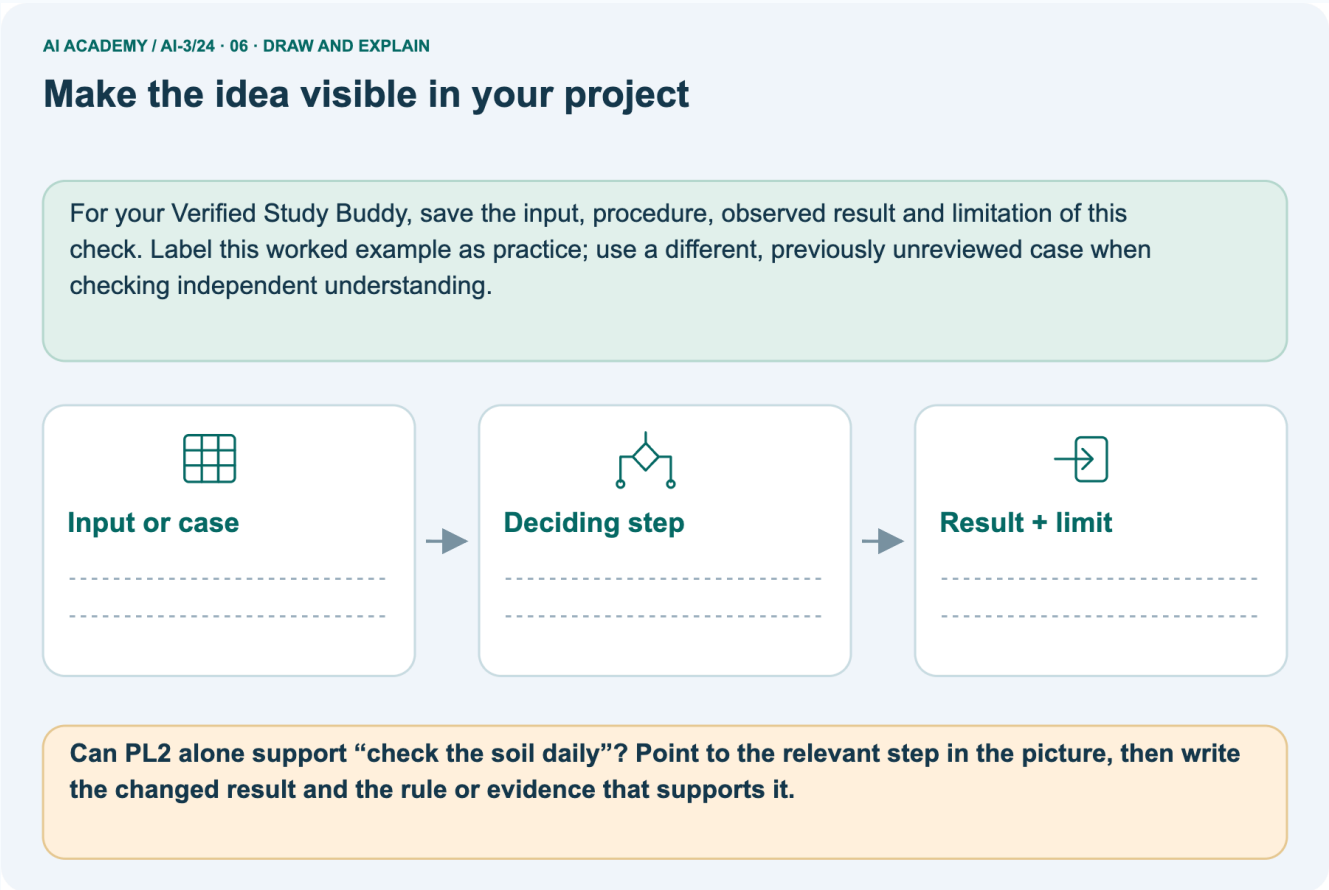
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A grounded **prototype** still needs an evaluation set and permission checks before anyone relies on it.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.